

Solving Exponential Equations using Logarithms - Nathaniel

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Student

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Topic: Solving Exponential Equations using Logarithms

Status: completed

Lesson Plan

Learning Objectives

- Solve exponential equations with different bases by applying logarithmic transformations to both sides.
- Calculate exact solutions for exponents using natural logarithms ($\ln$) and common logarithms ($\log$).
- Rearrange complex algebraic expressions involving logarithms to isolate the target variable.
- Model real-world growth and decay scenarios by constructing and solving exponential equations.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Great to see you. Before we dive into solving complex equations, let's flex those brain muscles with a quick recap. On the board, I've posted 4 'Quick Fire' tasks. For the first two on Change of Base: How would you rewrite $\log_5(20)$ using natural logs? And what is the numerical value of $\log_8(32)$? Now, for the Properties section: Can you expand $\ln(x^3/y)$ for me? Finally, condense $2\log(x) + \log(y)$ into a single term. Type your answers in the chat or write them directly on the screen!

Activities:

- *Quick Fire: Change of Base evaluation ($\log_5(20)$ and $\log_8(32)$)*
- *Quick Fire: Simplify properties (Expansion and Condensing)*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Imagine you are developing a new tech startup. Your user base is growing at 15% per month. If you start with 1,000 users, when exactly will you hit 1,000,000? We know the formula is $1000(1.15)^t = 1000000$. But the variable 't' is stuck in the ceiling! Today, we use logarithms as the 'ladder' to bring that variable down and solve for time precisely.

Activities:

- *Scenario visualization: Growth of a tech platform.*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Remember the most important rule for today: the Power Property. $\log(M^k) = k * \log(M)$. This is our secret weapon. If we have $3^x = 20$, and we take the log of both sides, that 'x' can jump down to the front. Nathaniel, tell me: if I have $\log(5^x)$, how can I rewrite it using this property?

Activities:

- *Re-stating the Power Property of Logarithms.*
- *Demonstrating 'Taking the Log of Both Sides'.*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at a 'Hard' level problem: $5^{(x+2)} = 3^{(2x)}$. Notice the bases are 5 and 3—no way to make them match easily. Step 1: Take the natural log (ln) of both sides. Step 2: Use the power property to bring the exponents down. Step 3: Distribute. Step 4: Collect all 'x' terms on one side. This is where your algebra skills shine. Let's walk through the distribution phase together.

Activities:

- *Step-by-step breakdown of equations with variables on both sides.*
- *Demonstration of isolating x: $(x+2)\ln 5 = 2x \ln 3$.*

5. Guided Practice (15 minutes)

Teacher Script:

Now your turn to drive. Solve $7^{(x-1)} = 2^x$. I want you to write each step on the whiteboard. Don't worry about the final decimal yet; let's find the 'Exact Solution' first. Remember to treat 'ln 7' just like a 2 or a 3—it's just a number! If you get stuck on the factoring step, remember we want 'x' all by itself.

Activities:

- *Student solves $7^{(x-1)} = 2^x$ step-by-step.*
- *Teacher provides real-time scaffolding for factoring out x.*

6. Independent Practice (15 minutes)

Teacher Script:

You're doing great. Here are three challenges. 1) Solve $4e^{(2x)} = 12$. 2) Solve $10^{(2x-3)} = 7^{(x+1)}$. 3) A strategy game calculation: If a unit's power doubles every 3 levels, at what level (x) will a power of 5 become 200? solve: $5 * 2^{(x/3)} = 200$. Go for it!

Activities:

- Solving equations with 'e' (natural base).
- Solving equations with different bases.
- Contextual problem solving.

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Today you mastered the 'Ladder' technique—using logs to bring down exponents. You handled different bases and even isolated variables using factoring! Before we log off, remember to join <https://learn.algonova.id/login> to access your guided notes. What was the trickiest part of isolating x today?

Activities:

- Summary of key steps: Log -> Power Rule -> Algebra -> Isolate.
- Login reminder.

Homework

Medium Questions:

1. $3^x = 55$
2. $8^{(x+1)} = 120$
3. $e^{(2x)} = 15$
4. $2 * 5^x = 80$

Hard Questions:

1. $4^{(x-2)} = 3^{(x+4)}$
2. $2^{(3x-1)} = 5^{(x+2)}$
3. $6^x = 2^{(x+1)} * 3^{(x-1)}$
4. $(1/2)^x = 7^{(x-3)}$
5. Solve for x: $5 * e^{(0.5x)} = 20 * 2^x$
6. Contextual: A population grows by $P = 500(1.04)^t$. Exactly when will it reach 1250? (Use logs for exact form).

Answer Key:

Medium: (1) $x = \log 55 / \log 3$ approx 3.65; (2) $x = (\log 120 / \log 8) - 1$ approx 1.30; (3) $x = \ln 15 / 2$ approx 1.35; (4) $x = \log 40 / \log 5$ approx 2.29

Hard: (1) $x = (2 \log 4 + 4 \log 3) / (\log 4 - \log 3)$ approx 25.4; (2) $x = (2 \log 5 + \log 2) / (3 \log 2 - \log 5)$ approx 8.35; (3) Use logs: $x \log 6 = (x+1)\log 2 + (x-1)\log 3 \rightarrow x = (\log 2 - \log 3) / (\log 6 - \log 2 - \log 3)$... since $\log 6 = \log 2 + \log 3$, this has unique properties/no solution or infinite check; (4) $x = (3 \log 7) / (\log 7 + \log 2)$ approx 1.95; (5) take ln: $\ln 5 + 0.5x = \ln 20 + x \ln 2$; (6) $t = \log(2.5) / \log(1.04)$ approx 23.36

REVIEW: Change of Base Formula & Evaluation - Nathaniel Juan Putra Winata Eng - Medium 1: evaluate $\log_4(10) \rightarrow \ln(10)/\ln(4)$ approx 1.66

REVIEW: Change of Base Formula & Evaluation - Nathaniel Juan Putra Winata Eng - Medium 2: evaluate $\log_2(30) \rightarrow \log(30)/\log(2)$ approx 4.91

REVIEW: Change of Base Formula & Evaluation - Nathaniel Juan Putra Winata Eng - Hard 1: Solve $\log_x(16) = 2/3 \rightarrow x^{(2/3)} = 16 \rightarrow x = 16^{(3/2)} = 64$

REVIEW: Change of Base Formula & Evaluation - Nathaniel Juan Putra Winata Eng - Hard 2: Simplify $(\log_3 5)(\log_5 27) \rightarrow (\log 5 / \log 3) * (\log 27 / \log 5) = \log 27 / \log 3 = 3$

REVIEW: Properties of Logarithms - Nathaniel Juan Putra Winata Eng - Medium 1: Expand $\log(x^2 y^3) \rightarrow 2\log x + 3\log y$

REVIEW: Properties of Logarithms - Nathaniel Juan Putra Winata Eng - Medium 2: Condense $3\ln(x) - \ln(y) \rightarrow \ln(x^3/y)$

REVIEW: Properties of Logarithms - Nathaniel Juan Putra Winata Eng - Hard 1: Expand $\log[\sqrt{x} / z^4] \rightarrow 1/2 \log x - 4 \log z$

REVIEW: Properties of Logarithms - Nathaniel Juan Putra Winata Eng - Hard 2: Solve $\log(x) + \log(x-3) = 1 \rightarrow \log(x^2 - 3x) = 1 \rightarrow x^2 - 3x = 10 \rightarrow (x-5)(x+2)=0 \rightarrow x=5$

Parent Reminder

Nathaniel's Math Progress: Solving the Unsolvable!

Today Nathaniel tackled 'Hard' level exponential equations. These are equations where common bases cannot be found, requiring him to use logarithms to 'unlock' variables from exponents. He did an excellent job navigating the complex algebraic steps required to isolate x . He's now able to model real-world growth precisely!